

Aman Sharma

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SUMMARY

AI engineer with a statistics foundation, shipping LLM products end-to-end — multi-agent orchestration, RAG, and fine-tuning, backed by guardrails and evaluation rigor that hold up in production.

EDUCATION

Indian Institute of Information Technology, Lucknow

Expected August 2027

M.Sc. Data Science | SGPA: 8.65/10

EXPERIENCE

AI Agent Developer Intern

May 2026 – June 2026

Genotek Global

Remote

- Built and deployed a **Telegram bot** on **AWS EC2** for field sales — GPS-verified check-in/check-out with a 3-gate anti-cheat system (live location · freshness · accuracy · Haversine distance), Google Places verification, **Supabase** backend, and HubSpot CRM sync; validated end-to-end with company leadership
- Shipped an LLM meeting-debrief pipeline — local **faster-whisper** (CPU int8, Hinglish-aware) → **Groq LLaMA 3.1 70B** → structured HubSpot engagements — atop a **15-state python-telegram-bot FSM** with multi-rep session isolation and drift classification (clean/suspicious/flagged) + manager alerting

Data Science Intern

Sep 2025 – Nov 2025

Climate Resilience Observatory (CRO), IIIT Lucknow

Lucknow, India

- Conducted EDA on live meteorological data from NASA POWER and ECMWF ERA5 to identify precipitation patterns across 72 Uttar Pradesh districts
- Performed model selection and analytical evaluation for binary rainfall forecasting in collaboration with the Office of the Relief Commissioner, Govt. of Uttar Pradesh

PROJECTS

WealthOS — Multi-Agent Financial Intelligence Platform 

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LangGraph, FastAPI, Qdrant, DSPy, E2B, LangSmith

- Orchestrated **8 agents** via a LangGraph 8-node state machine (**45 tools across 7 MCP servers** — market data, SEC, news, tax, portfolio); **asyncio.gather** parallelism delivers an investment memo in **60–90s** (~2× over sequential) on Groq llama-3.3-70b-versatile, with Pydantic-validated outputs and PII-masked LangSmith traces for auditability
- Built a hybrid **Agentic RAG** over SEC 10-K filings in Qdrant (all-MiniLM dense + BM25 sparse via reciprocal-rank fusion, Cohere rerank, \$0 embedding cost); added **DSPy**-compiled prompts, **E2B**-sandboxed DCF + Monte Carlo, and a 5-dim Financial Health Score — traced in LangSmith + W&B Weave

Resolv — Guardrail Refund Agent & pass^k Reliability Benchmark 

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Google ADK, litellm, TRL (GRPO/RLVR)

- Built a harness-first refund agent on **Google ADK** + provider-agnostic **litellm**: **2 LLM stages** (extract, resolve) only *propose*, while **7 first-match policy rules** in deterministic Python are the sole write path, vetoing every payout before money moves; refund caps derive from the order record — not the customer's stated amount — so inflated claims are refused by arithmetic
- τ -**bench-style pass^k** eval vs an adversarial LLM customer — **pass³ 0.96, 0 unauthorized refunds / 200 runs**; ablation: harness *off* leaked **\$2,120** vs **\$0 on**; **RLVR**-tuned the extractor (TRL GRPO, Qwen2.5-1.5B LoRA) on the production verifier (**no LLM judge**), **0.49→0.71** on 92 held-out; **240 tests** + regression-gated CI + Docker

RetailGraph — Multimodal Product Knowledge Graph 

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Qwen2-VL-7B, QLoRA, GraphRAG, Neo4j, Groq, RAGAS

- Fine-tuned **Qwen2-VL-7B** (QLoRA, 4-bit) on **5,369 verified pairs** (Modal A100), hitting **94.2% exact-match** across 11 fields with 0 parse errors via schema-validated retry; served through a **6-node LangGraph agent** routing queries to Cypher, GraphRAG (Qdrant + Neo4j), or analytics on FastAPI + Groq Llama-3.3-70B
- Benchmarked **GraphRAG vs vector-only** on 20 queries — vector scored **0/20** on constraints — built a **30-query eval harness** scoring route/intent/Cypher **execution accuracy (96.7% / 93.3% / 84.2%)** against live Neo4j, plus **RAGAS** (Groq-judged) and **LLM-as-Judge (GPT-4o-mini)** surfacing 38 hits exact-match missed

SKILLS

Agentic AI: Google ADK, MCP, LangGraph, LangChain, Agentic RAG, GraphRAG, DSPy, Mem0, Policy Engines & Guardrails
LLM & Fine-tuning: SFT, ORPO, GRPO, RLVR, QLoRA, LoRA, LlamaFactory, HuggingFace, LLM-as-Judge, Ragas, Hybrid Search, Prompt Engineering, Vision-Language Models (Qwen2-VL)

Languages: Python (PyTorch, NumPy, Pandas, Scikit-learn, Matplotlib), C, SQL

Mathematics & Statistics: Linear Algebra, Calculus, Probability Theory, Statistical Inference, Hypothesis Testing

Machine Learning: Transformers, Weak Supervision (Snorkel), Supervised & Unsupervised Learning, CNN, LSTM

Databases & Search: Neo4j (Cypher), Qdrant, pgvector, FAISS, Redis, PostgreSQL

Developer Tools: AWS, Docker, FastAPI, Streamlit, LangSmith, W&B, Modal, MLflow

ACHIEVEMENTS

- IIT JAM MS:** Secured All India Rank **219** in IIT JAM 2025 (Mathematical Statistics)
- GATE Statistics:** Secured All India Rank **493** in GATE 2025